

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>
let's denote the initial number of vacuum cleaners Melanie had as x
at the green house, she sold a third of the vacuum cleaners, which is $x/3$
at the red house, she sold 2 more vacuum cleaners
at the orange house, she sold half of what was left, which is $(x - x/3 - 2)/2$
the total number of vacuum cleaners she sold is $x/3 + 2 + (x - x/3 - 2)/2$
simplifying the expression, we get $x/3 + 2 + (2x - 2)/2$
further simplifying, we get $x/3 + 2 + x - 1$
combining like terms, we get $x/3 + x + 1$
since she had 5 vacuum cleaners left, the total number of vacuum cleaners she sold is $x - 5$
equating the two expressions, we get $x/3 + x + 1 = x - 5$
combining like terms, we get $4x/3 = 6$
multiplying both sides by $3/4$, we get $x = 9/2$
 $x = 4.5$
</think><answer>
4.5
</answer>